A Statistical Analysis of doctor visit counts using Poisson Regression and Bootstrap Methods

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1 Introduction

We are looking at data collected concerning the number of doctor visits to a family doctor for a sample of individuals in the US in 2001. The study specifically excludes people receiving public insurance and only focuses only people working in the private sector (not including people who identify as self-employed).

This report aims to investigate how demographic and socioeconomic variables—such as age, income, education, sex, marital status, ethnicity, health insurance coverage, and chronic illness—relate to the frequency of doctor visits. Using generalised linear models (GLMs) with a Poisson specification, along with sandwich estimators and bootstrap resampling methods, the analysis aims to draw statistical inferences by estimating effects, constructing confidence intervals, and testing hypotheses with improved accuracy and robustness.

2 Exploratory Data Analysis

We will initially look at simple summary statistics and plots to motivate our approaches and methodologies for the statistical analysis techniques.

Table 1: Summary Statistics of Variables in `dvisus.csv`

Variables	**N**	**Mean**	**SD**	**Min**	**25%**	**Max**
Doctor visits (`docvis`)	1084	3.39	4.95	0.00	0.00	31.00
Age (in decades) (`age`)	1084	4.16	1.00	2.50	3.30	6.40
Years of schooling (`educ`)	1084	13.10	2.49	6.00	12.00	17.00
Log income (/1000) (`lincome`)	1084	3.45	0.60	2.27	3.00	5.22
Female (indicator =1) (`female`)	1084	0.47	0.50	0.00	0.00	1.00
Married (indicator =1) (`married`)	1084	0.67	0.47	0.00	0.00	1.00
White (indicator =1) (`white`)	1084	0.69	0.46	0.00	0.00	1.00
Hispanic (indicator =1) (`hispanic`)	1084	0.19	0.39	0.00	0.00	1.00
Private insurance (indicator =1) (`private`)	1084	0.87	0.33	0.00	1.00	1.00
Chronic condition (indicator =1) (`chronic`)	1084	0.34	0.47	0.00	0.00	1.00

Figure 1a in addition to the summary statistics from Table 1 reveals substantial variation in doctor visit frequency, with a large number reporting zero visits in the year prior and a long right tail indicating a strong skewness. This distribution makes the Poisson GLM an appropriate modelling choice, as it is designed for count data, handles skewness, and links the expected count to covariates through a log function. Furthermore, the summary suggests overdispersion as the variance of the data (24.50) exceeds its mean (3.39) which violates the core assumption of the standard Poisson model which could lead to inaccurate standard errors. This reinforces the need for robust standard errors or bootstrap methods to ensure reliable inference.
The distribution of age is reasonably uniform across the working-age adults (Figure 1b) while log-income

is slightly skewed (Figure 1c), indicating income inequality within the sample. Education levels (Figure 1d) are concentrated around 12 to 16 years, suggesting that most individuals have completed high school or some college. These patterns suggest potential associations between socioeconomic status, health status, and healthcare utilisation. This further motivates the need for a formal regression analysis to quantify these effects and make robust inferences.

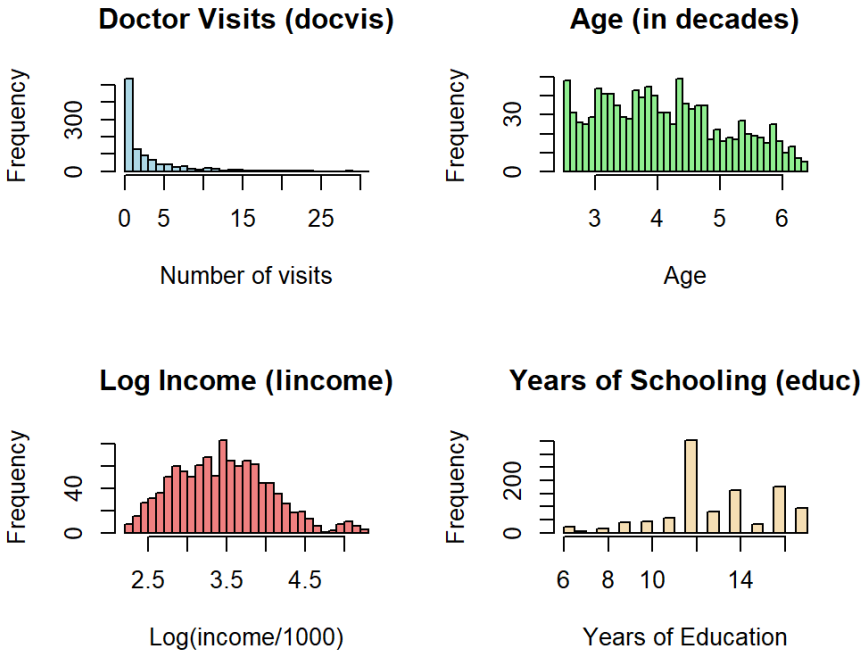


Figure 1: Relevant univariate plots (a-d from left to right, top to bottom)

3 Estimation and computation of standard errors

Motivated from the univariate plots and the summary statistics, the model we will use for the distribution of doctor visits is a Poisson GLM. The canonical link we will use the log link whereby:

$$\ln(\mu) = \eta = x^\intercal \beta \implies \mu = \exp(x^\intercal \beta)$$

which gives an easy multiplicative interpretation of the coefficients (percentage change in expected count with respect to a unit increase in variable). Hence the Poisson GLM used is:

```
mod_pois <- glm(formula = docvis ~ age + lincome + female + educ + married +
white + hispanic + private + chronic, family = poisson(link = "log"),
data = dvisus)
```

More importantly, for this Poisson GLM with the canonical link function, the estimated model specifications and standard errors are estimated now. However, as mentioned in the exploratory data analysis section, there is an evident overdispersion. Hence we will use the Sandwich Estimator to obtain an estimate for the variance of the parameter estimates that are robust to general forms of heteroscedasticity (in this case overdispersion). Here the HC0 (White Estimator) used, to achieve a baseline robust standard errors without further adjustments with consideration to leverage and outliers. The table below demonstrates this information:

```
# Table 2: Estimates with standard and robust (sandwich) standard errors

             Estimate  Std_SE  Robust_SE
(Intercept) -0.95300  0.1577  0.43239
age          0.07288  0.0171  0.04266
lincome      0.05846  0.0307  0.07583
female       0.58299  0.0353  0.08687
educ         0.02845  0.0083  0.01975
married      0.09476  0.0371  0.09488
white        0.20261  0.0545  0.13526
hispanic    -0.08659  0.0740  0.17600
private      0.43375  0.0725  0.18131
chronic      0.79083  0.0342  0.08379
```

The comparison between the conventional and robust standard errors reveals noticeable differences for several coefficients (e.g. married, white, hispanic, private). Particularly, the robust standard errors are all larger than the standard Poisson GLM errors which aligns with the earlier evidence of overdispersion in the data. Larger robust standard errors indicate that the model-based Poisson assumption massively underestimates the uncertainty of our estimates which can lead to overconfident inference (e.g. p-values that are too small or tight confidence intervals). The sandwich estimator corrects this by adjusting the variance-covariance matrix by not assuming constant error variance and instead directly estimates variances from observed residuals. This is all done while not altering the coefficient estimates themselves.

As a result, inference (e.g. significance testing) based on robust (sandwich) standard errors is more reliable in this setting and should be preferred over standard Poisson-based inference when overdispersion is present.

4 Bootstrap

We would like to estimate standard errors without relying on Poisson model and its assumption, particularly that the variance is equal to the mean. Considering we observed overdispersion, the bootstrap provides a data-driven, nonparametric way to determine the variability of the coefficient estimates to make our inferences more reliable and robust. Bootstrap achieves this by resampling from the observed data in order to approximate the sampling distribution of the estimator. The methodology behind this is:

1. Repeatedly drawing random samples with replacement from the original dataset
2. For each resample, it recomputes the estimator (e.g. the regression coefficients)
3. The variation across these estimates mimics how the estimator would vary if we had many samples from the true population
4. This allows us to estimate the standard error, bias and even confidence intervals without having to rely on assumptions about the underlying distribution or even the variance structure.

Applying this to the current data with 1000 resamples we get standard errors:

```
# Table 3: Estimates with conventional, robust, and bootstrap standard errors
             Estimate  Std_SE Robust_SE Bootstrap_SE
(Intercept) -0.95300 0.15770   0.43239      0.44730
age          0.07288 0.01714   0.04266      0.04259
lincome      0.05846 0.03074   0.07583      0.07547
female       0.58299 0.03527   0.08687      0.08539
educ         0.02845 0.00827   0.01975      0.02035
married      0.09476 0.03706   0.09488      0.09811
white        0.20261 0.05447   0.13526      0.13732
hispanic    -0.08659 0.07395   0.17600      0.17522
private      0.43375 0.07249   0.18131      0.18079
chronic      0.79083 0.03417   0.08379      0.08496
```

In general, the bootstrap standard errors are similar in magnitude to the robust (sandwich) standard errors, and both are often larger than the standard Poisson model errors. This reinforces the earlier finding of

overdispersion, suggesting that model-based standard errors underestimate true variability. The bootstrap provides an additional nonparametric, data-driven check. This confirms the need for robust inference methods in this context to get trustworthy and not too optimistic results.

5 Wald test

Now we wish to test the hypothesis:

$$H_0 : \beta_{\text{lincome}} = \beta_{\text{married}} = \beta_{\text{hispanic}} = 0 \quad \textbf{vs} \quad H_1 : \text{at least one of them is non-zero}$$

using the covariance matrix derived from the Poisson GLM, the sandwich, and bootstrap standard errors. To carry out this hypothesis test, we must use a Wald test which uses the Wald Statistic:

$$W = \hat{\beta}^{\intercal}[\text{Var}(\hat{\beta})]^{-1}\hat{\beta}$$

where the inverse covariance matrix would differ depending on each of the Poisson GLM, the sandwich, and bootstrap standard errors. Under H_0, we have that the Wald statistic follows a chi-squared distribution with degrees of freedom equal to the number of tested parameters. In this case:

$$W_{\text{pois}}, W_{\text{sand}}, W_{\text{boot}} \sim \chi^2_3$$

Using this, p-values can be calculated by looking at the probability of observing an event at least as extreme as this under H_0. The final results are:

```
# Table 4: The respective Wald statistics and the resulting p-values of the test
    described above

      Method     Statistic      P_Value
      Standard    10.42749      0.01526
      Sandwich     1.58634      0.66249
      Bootstrap    1.52436      0.67666
```

The Wald test evaluates whether the combined effects of lincome, married, and hispanic are jointly zero.

- Standard Poisson GLM gives a lower p-value of 1.5 percent. This is lower than the typical significance level of 5 percent. This means, under the Poisson GLM, there is sufficient evidence to reject H_0 indicating that the combined effects of lincome, married, and hispanic are not jointly zero
- Sandwich gives a much higher p-value of 66 percent. This is much higher than the typical significance level of 5 percent. This means, under the robust (sandwich) standard errors, there is insufficient evidence to reject the H_0
- Bootstrap gives a much higher p-value of 68 percent. This is much higher than the typical significance level of 5 percent. This means, under the bootstrap standard errors, there is insufficient evidence to reject the H_0

This reinforces the earlier findings that the model-based inference may be far too optimistic. This demonstrates the importance of using robust methods for joint hypothesis under overdispersion (or other general forms of heteroscedasticity).

6 Bootstrap p-value

Now we will compute a bootstrap-based (empirical) p-value specifically for Wald Statistic derived from the sandwich covariance matrix. So simulated recomposed Wald Statistics are generated then the p-value is obtained by looking at what proportion of those simulated Wald Statistics are equal or greater than the observed Wald Statistics. For this example 1000 simulations are used, and the achieved p-value is 0.798 which is larger than the p-values calculated previously analytically. Unlike the analytic chi-squared p-value, this approach makes no assumptions about the test statistic's distribution and accounts for model uncertainty more fully. If the bootstrap p-value is noticeably higher, it suggests that the analytic p-value may be giving a false sense of confidence in results which is stemming from the assumption that Wald Statistic perfectly follows a chi-square distribution.

7 Estimation results and Studentised confidence intervals

Imposing on the restriction $\beta_{\text{lincome}} = \beta_{\text{married}} = \beta_{\text{hispanic}} = 0$ we end with a resulting model:

```
mod_restricted <- glm(docvis ~ age + female + educ + white + private + chronic,
                      family = poisson(link = "log"),
                      data = data)
```

In order to continue and calculate studentised confidence intervals, we need the studentised statistic. A studentised statistic is an estimate divided by its estimated standard error. For a parameter θ, the studentised form is:

$$T = \frac{\hat{\theta} - \theta}{\hat{\sigma}}$$

In the bootstrap context, we do not know the true value θ, so we approximate its distribution using resampled estimates. The corresponding studentised bootstrap statistic is:

$$T^* = \frac{\hat{\theta}^* - \hat{\theta}}{\hat{\sigma}^*}$$

- $\hat{\theta}^*$ is the estimate from a bootstrap sample,
- $\hat{\theta}$ is the estimate from the original full sample,
- $\hat{\sigma}^*$ is the standard error computed from the bootstrap sample.

This approach standardises each bootstrap estimate by its own variability, producing a more stable and often more accurate distribution for constructing confidence intervals so we will use it. Once again we're using 1000 simulations and an $\alpha = 0.05$ for the typical 95 percent confidence interval. Moreover, using the canonical link function, we can translate the coefficient estimates into understood results.

Coefficient	Estimate	Multiplicative_Effect	CI_Lower	CI_Upper
(Intercept)	-0.9036	0.4051	0.2121	0.7516
age	0.0803	1.0836	0.9924	1.1792
female	0.5700	1.7683	1.5075	2.0972
educ	0.0357	1.0364	0.9989	1.0727
white	0.2610	1.2983	1.0798	1.5526
private	0.4787	1.6140	1.1648	2.3252
chronic	0.7945	2.2134	1.8761	2.6216

To put the multiplicative effect in layman's terms, the results state:

- Females are expected to have about 1.77 times as many doctor visits as their male counterparts. A 76.8 percent increased chance holding other variables constant.
- Chronic conditions have the largest impact on the number of doctor visits : individuals with a chronic illness have about 2.21 times as many visits as those without. A 121 percent increased chance holding other variables constant.
- Private insurance increases the expected visits by 61 percent holding other variables constant.
- White individuals have about 30 percent more visits than non-white individuals, holding other variables constant.
- Education has a smaller and borderline effect: every additional year of education increases visits by about 3.6 percent holding other variables constant.
- Age also has a modest effect: each additional decade of age increases expected visits by about 8.4 percent holding other variables constant.

Appendix

```r
# Load required packages
library(ggplot2)
library(sandwich)
library(lmtest)
library(boot)
# Read data
data <- read.csv(file.choose())
##1. Summary statistics for all variables
summary(data)
##2. Univariate plots for selected variables
# Set up 2x2 plot lay
par(mfrow = c(2, 2))
# Histogram for docvis
hist(data$docvis,
     main = "Doctor Visits (docvis)",
     xlab = "Number of visits",
     col = "lightblue",
     breaks = 30)
# Histogram for age
hist(data$age,
     main = "Age (in decades)",
     xlab = "Age",
     col = "lightgreen",
     breaks = 30)
# Histogram for lincome
hist(data$lincome,
     main = "Log Income (lincome)",
     xlab = "Log(income/1000)",
     col = "lightcoral",
     breaks = 30)
# Histogram for educ
hist(data$educ,
     main = "Years of Schooling (educ)",
     xlab = "Years of Education",
     col = "wheat",
     breaks = 30)
# Reset layout
par(mfrow = c(1, 1))
##2 Estimation and Computation of standard errors
  #Fit Poisson GLM with canonical (log) link
# Model includes all covariates: age to chronic (no interactions)
mod_pois <- glm(docvis ~ age + lincome + female + educ + married + white + hispanic
    + private + chronic,
                data = data,
                family = poisson(link = "log"))

  #Standard summary with conventional standard errors
summary(mod_pois)

  #Compute sandwich (robust) standard errors
# HC0 = classic White estimator
robust_vcov <- vcovHC(mod_pois, type = "HC0")

  #Display coefficients with robust SEs
coeftest(mod_pois, vcov. = robust_vcov)

#Compare standard errors side-by-side
standard_se <- sqrt(diag(vcov(mod_pois)))
robust_se <- sqrt(diag(robust_vcov))
```

```r
se_comparison <- data.frame(
  Estimate = coef(mod_pois),
  Std_SE = standard_se,
  Robust_SE = robust_se
)

print(se_comparison, digits = 4)

##4 Bootstrap
# Define a function to refit the Poisson model on a bootstrap resample
bootstrap_fn <- function(data, indices) {
  d <- data[indices, ]  # paired resampling (rows)
  fit <- glm(docvis ~ age + lincome + female + educ + married +
               white + hispanic + private + chronic,
             family = poisson(link = "log"),
             data = d)
  return(coef(fit))  # return estimated coefficients
}
# Set seed for reproducibility
set.seed(42069)
# Run the bootstrap
boot_results <- boot(data = data, statistic = bootstrap_fn, R = 1000)
# Compute bootstrap standard errors
boot_se <- apply(boot_results$t, 2, sd)
# Create a comparison table
bootstrap_comparison <- data.frame(
  Estimate = coef(mod_pois),
  Std_SE = sqrt(diag(vcov(mod_pois))),
  Robust_SE = sqrt(diag(vcovHC(mod_pois, type = "HC0"))),
  Bootstrap_SE = boot_se
)
# Print rounded results
print(round(bootstrap_comparison, 5))
##5 Wald test
# Define the vector of coefficients to test
params <- c("lincome", "married", "hispanic")
# ExtracT coefficient estimates
beta_hat <- coef(mod_pois)[params]
# Standard variance matrix
V_standard <- vcov(mod_pois)[params, params]
# Sandwich (robust) variance matrix
V_robust <- vcovHC(mod_pois, type = "HC0")[params, params]
# Bootstrap variance matrix
# (From previous Q3 bootstrap results)
colnames(boot_results$t) <- names(coef(mod_pois))
V_bootstrap <- cov(boot_results$t[, params])
# Function to compute Wald statistic and p-value
wald_test <- function(beta, V, df = length(beta)) {
  W <- t(beta) %*% solve(V) %*% beta
  pval <- 1 - pchisq(W, df)
  return(c(statistic = W, p_value = pval))
}
# Compute test results
wald_standard <- wald_test(beta_hat, V_standard)
wald_robust   <- wald_test(beta_hat, V_robust)
wald_boot     <- wald_test(beta_hat, V_bootstrap)
# Combine into a summary table
wald_summary <- data.frame(
  Method = c("Standard", "Sandwich (Robust)", "Bootstrap"),
  Statistic = round(c(wald_standard[1], wald_robust[1], wald_boot[1]), 5),
  P_Value  = round(c(wald_standard[2], wald_robust[2], wald_boot[2]), 5)
```

```
)
print(wald_summary)
##6 Bootstrap p-value
# Function to compute Wald statistic on a bootstrap sample
wald_boot_fn <- function(data, indices) {
  d <- data[indices, ]
  fit <- glm(docvis ~ age + lincome + female + educ + married +
               white + hispanic + private + chronic,
             family = poisson(link = "log"),
             data = d)
  # Extract coefficients and sandwich variance
  bhat <- coef(fit)[params]
  Vhat <- vcovHC(fit, type = "HC0")[params, params]
  W <- t(bhat) %*% solve(Vhat) %*% bhat
  return(as.numeric(W))
}
# Run bootstrap for Wald statistic
set.seed(69420)
wald_boot_dist <- boot(data = data, statistic = wald_boot_fn, R = 1000)
# Observed Wald stat from Q4 using sandwich variance
wald_obs <- as.numeric(wald_test(beta_hat, V_robust)[1])
# Compute bootstrap p-value
pval_bootstrap <- mean(wald_boot_dist$t >= wald_obs)
print(pval_bootstrap)
##7 Estimation Results and Studentised CI
# 1. Fit the restricted Poisson model
mod_restricted <- glm(docvis ~ age + female + educ + white + private + chronic,
                      family = poisson(link = "log"),
                      data = data)
# 2. Extract original estimates and standard errors
original_est <- coef(mod_restricted)
original_se  <- sqrt(diag(vcov(mod_restricted)))
# 3. Define studentized bootstrap function
studentized_fn <- function(data, indices) {
  d <- data[indices, ]
  fit <- glm(docvis ~ age + female + educ + white + private + chronic,
             family = poisson(link = "log"),
             data = d)
  est <- coef(fit)
  se  <- sqrt(diag(vcov(fit)))
  return((est - original_est) / se)  # studentized t-statistics
}
# 4. Run the bootstrap
set.seed(21)
boot_res <- boot(data = data, statistic = studentized_fn, R = 1000)
# 5. Compute 95% studentized confidence intervals
alpha <- 0.05
ci_lower <- numeric(length(original_est))
ci_upper <- numeric(length(original_est))
names(ci_lower) <- names(ci_upper) <- names(original_est)
for (j in seq_along(original_est)) {
  t_star <- boot_res$t[, j]
  t_star <- t_star[is.finite(t_star)]  # remove NaNs or Infs
  q_low  <- quantile(t_star, probs = 1 - alpha / 2)
  q_high <- quantile(t_star, probs = alpha / 2)

  ci_lower[j] <- original_est[j] - q_low * original_se[j]
  ci_upper[j] <- original_est[j] - q_high * original_se[j]
}
# 6. Compute multiplicative effects and format results
multiplicative_effects <- exp(original_est)
ci_exp_lower <- exp(ci_lower)
```

```
ci_exp_upper <- exp(ci_upper)
# 7. Summary table
studentized_ci_table <- data.frame(
  Coefficient = names(original_est),
  Estimate = round(original_est, 4),
  Multiplicative_Effect = round(multiplicative_effects, 4),
  CI_Lower = round(ci_exp_lower, 4),
  CI_Upper = round(ci_exp_upper, 4)
)
print(studentized_ci_table)
```